

TIA–FPGA – FPGA replacement for the Atari 2600 Television Interface Adaptor (C010444)

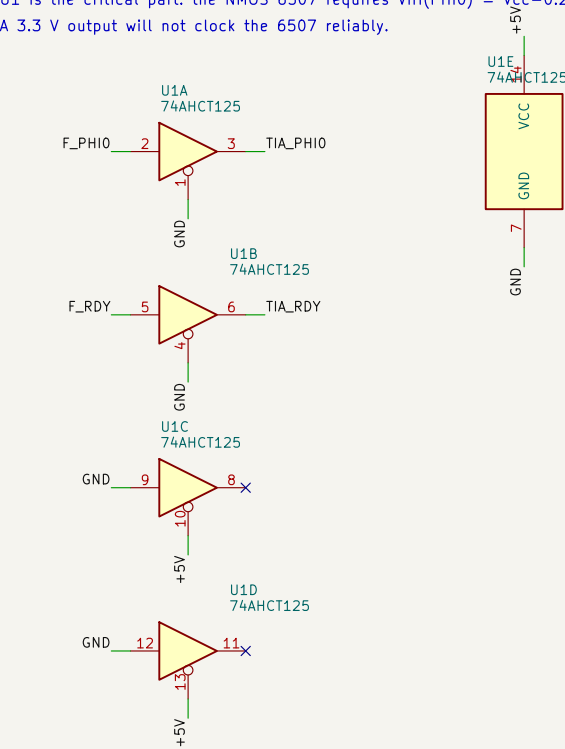
Rev A – digital interface only. Chroma and paddles do not fit the pin budget; see the notes.

FPGA MODULE – Sipeed Tang Nano 9K (GW1NR–LV9QN88PC6/I5)

Pin map verified against both Sipeed board revisions (3672 and 3674), which agree. Of the 48 header pins only 29 are usable 3.3 V GPIO: 8 are BANK3 at 1.8 V, 8 are HDMI differential pairs, 3 are power. Rev A uses all 29, with none spare.

LEVEL TRANSLATORS

U1 is the critical part: the NMOS 6507 requires $V_{IH}(PHIO) = V_{cc}-0.2\text{ V} \sim 4.8\text{ V}$. A 3.3 V output will not clock the 6507 reliably.

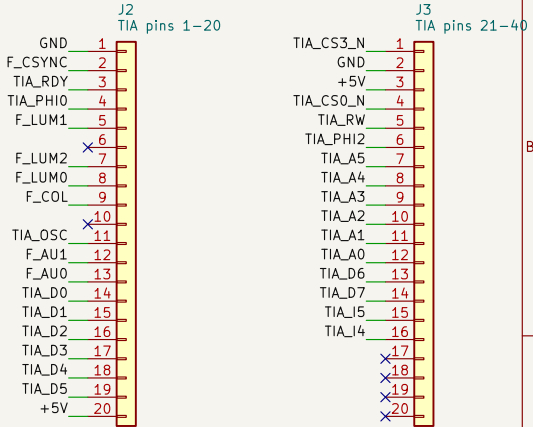
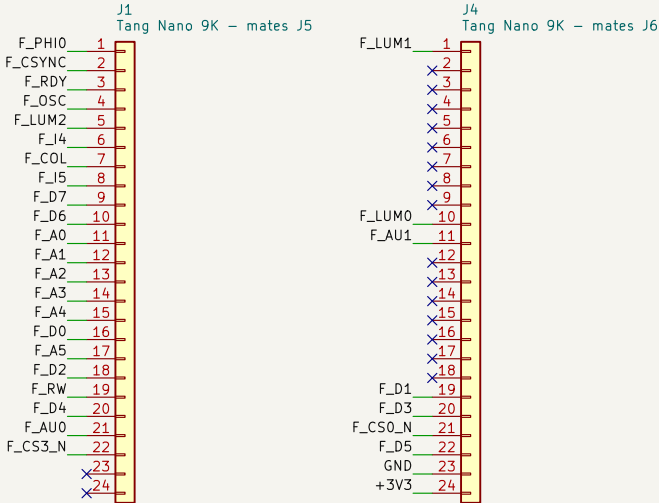


COLOUR VIDEO – via the Tang Nano’s HDMI connector

The 7-bit colour value never leaves the FPGA: an internal TMDS encoder drives the HDMI pins, which are dedicated differential pairs on the module and were never part of the 29-pin GPIO budget. Colour costs zero header pins. FPGA pins (P/N): CK 69/68, D0 71/70, D1 73/72, D2 75/74. RULE: leave J1/23–24 and J4/12–17 open – they carry the same TMDS pairs, and any stub degrades the signal.

TIA SOCKET – DIP–40

Two 20-way SIP strips that plug into the 2600 socket or a breadboard, in place of the C010444.

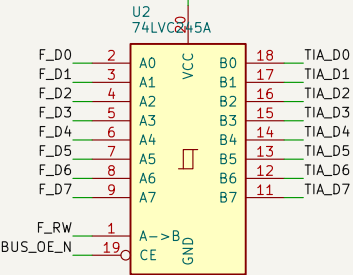


J4/2–9 are BANK3 at 1.8 V – unusable at 3.3 V without a local LDO and a 1.8 V-powered buffer. Recovering those 8 pins is the rev B fix.

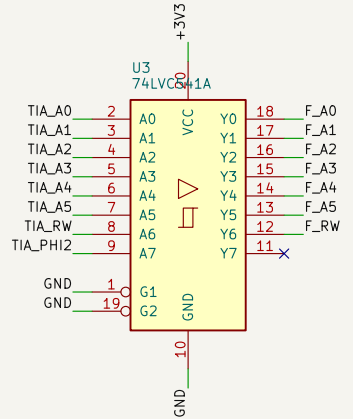
J4/18 (+5 V from the Tang Nano) left open on purpose: this board is fed from the breadboard rail, so tying both would put two sources on one net. Power the Tang Nano over USB. +3V3 is taken from J4/24.

U1 runs from +5 V – this is where the level step-up happens. Spare buffers 3 and 4: /OE tied to +5 V, input tied to GND.

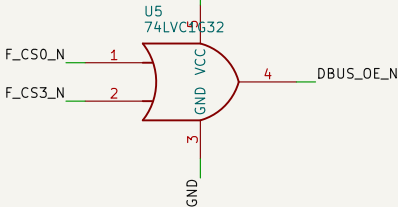
U2 D0–D7 bidirectional. A side = FPGA, B side = TIA. DIR comes straight from the buffered R/W: DIR=1 (a read) sends FPGA → TIA. /OE comes from U5, so neither 245 control line costs an FPGA pin – that freed the one we needed.



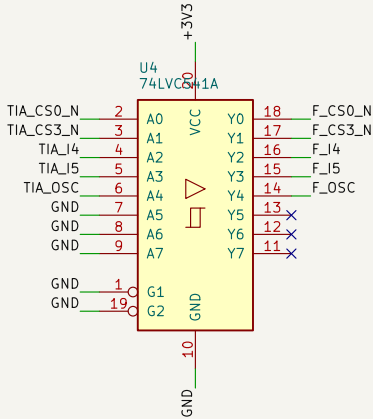
U3 A0–A5, R/W and PHIO – permanently enabled (G1=G2=GND).



U5 /OE = /CS0 OR /CS3. The 245 drives the bus only while the TIA is selected, so it never fights the ROM or the RIOT on a CPU read.



U4 /CS0, /CS3, triggers and the 3.579545 MHz clock from the crystal.



VIDEO OUTPUT – CX–2600A resistor network (STANDALONE USE ONLY)

4:2:1 luma weighting, plus R9 for chroma. Do NOT populate when the board goes into a real 2600: the console already has this network, and a second one in parallel would shift every level. Drive the socket pins and stop.

